

Task Name: Create a Java program to input two numbers and find the largest number.

Objective:

- ❖ To practice taking input using the Scanner class
- ❖ To understand decision-making statements in Java
- ❖ To learn number comparison techniques
- ❖ To identify the maximum value between two numbers
- ❖ To build basic Java programming logic

Problem Analysis:

In this task, the program will take two integer values from the user. After taking input, the program will check which number is greater using conditional statements. If both numbers are the same, the program will display an equal message. Otherwise, it will show the larger number.

Algorithm:

- Step 1: Begin the program
- Step 2: Declare variables
- Step 3: Input the first number
- Step 4: Input the second number
- Step 5: Check if the first number is greater
- Step 6: Check if the second number is greater
- Step 7: Otherwise, the numbers are equal
- Step 8: Display result
- Step 9: Stop

Source Code:

```
import java.util.Scanner;

public class Maxnumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Input first value: ");

        int a = sc.nextInt();

        System.out.print("Input second value: ");

        int b = sc.nextInt();

        if(a > b)

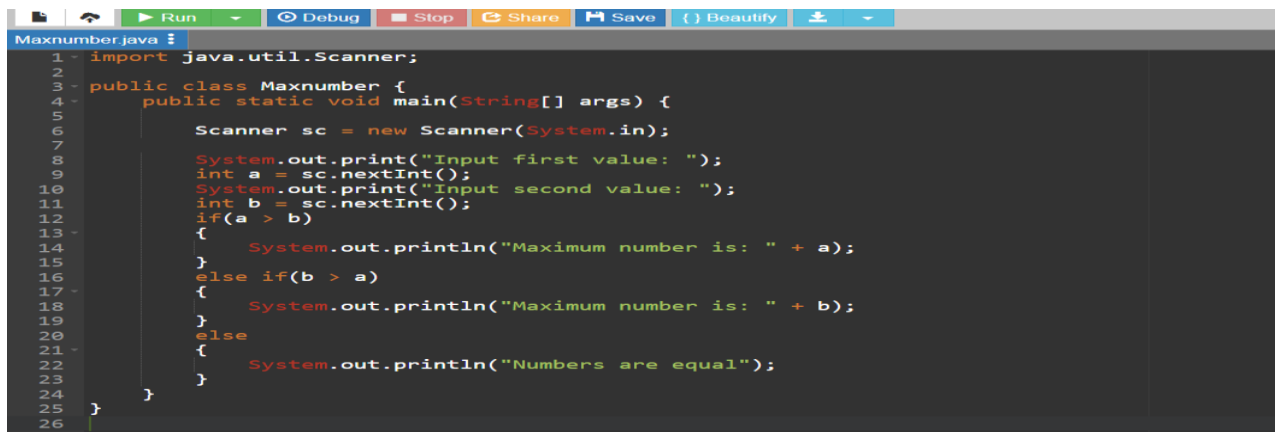
        {
```

```

        System.out.println("Maximum number is: " + a);
    }
    else if(b > a)
    {
        System.out.println("Maximum number is: " + b);
    }
    else
    {
        System.out.println("Numbers are equal");
    }
}
}

```

Input:

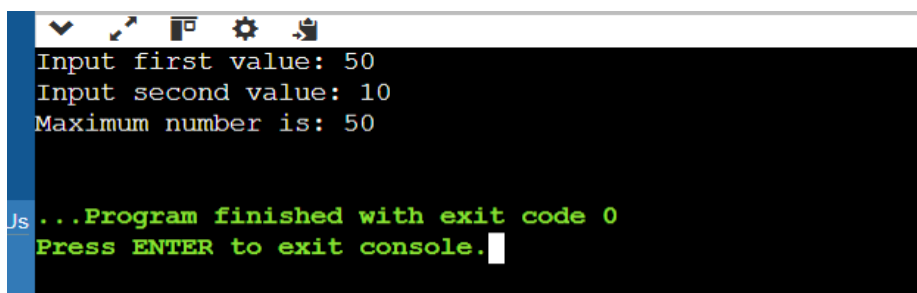


```

Maxnumber.java
1- import java.util.Scanner;
2-
3- public class Maxnumber {
4-     public static void main(String[] args) {
5-
6-         Scanner sc = new Scanner(System.in);
7-
8-         System.out.print("Input first value: ");
9-         int a = sc.nextInt();
10-        System.out.print("Input second value: ");
11-        int b = sc.nextInt();
12-        if(a > b)
13-        {
14-            System.out.println("Maximum number is: " + a);
15-        }
16-        else if(b > a)
17-        {
18-            System.out.println("Maximum number is: " + b);
19-        }
20-        else
21-        {
22-            System.out.println("Numbers are equal");
23-        }
24-    }
25- }
26-

```

Output:



```

Input first value: 50
Input second value: 10
Maximum number is: 50

...Program finished with exit code 0
Press ENTER to exit console.

```

Discussion & Conclusion: In this experiment, a Java program was developed to take two numbers as input from the user and determine the largest number between them. The program uses conditional statements (if-else) to compare the values. If the first number is greater than the second, it is displayed as the largest; otherwise, the second number is considered larger. This simple comparison helps in understanding the basic logic-building concept and decision-making process in Java programming. The experiment successfully demonstrates how to compare two numbers using Java. It improves understanding of user input, conditional statements, and program execution. This basic concept is essential for solving more complex problems in programming.

Task 2

Task Name: Develop a Java calculator program using switch case.

Objective:

- To understand how switch-case works
- To learn the implementation of arithmetic operations
- To practice operator-based decision making
- To create a simple calculator program
- To improve Java coding skills

Problem Analysis:

Here, the user will enter two numeric values and one mathematical operator. The program will check the operator using a switch statement and perform the selected calculation. Finally, the result will be displayed on the screen.

Algorithm:

Step 1: Start program
Step 2: Take first value
Step 3: Take second value
Step 4: Take operator symbol
Step 5: Apply switch statement
Step 6: Perform operation based on operator
Step 7: Show calculated result
Step 8: End program

Source Code:

```
import java.util.Scanner;

public class SimpleCalculator {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter value 1: ");

        double x = sc.nextDouble();

        System.out.print("Enter value 2: ");

        double y = sc.nextDouble();
```

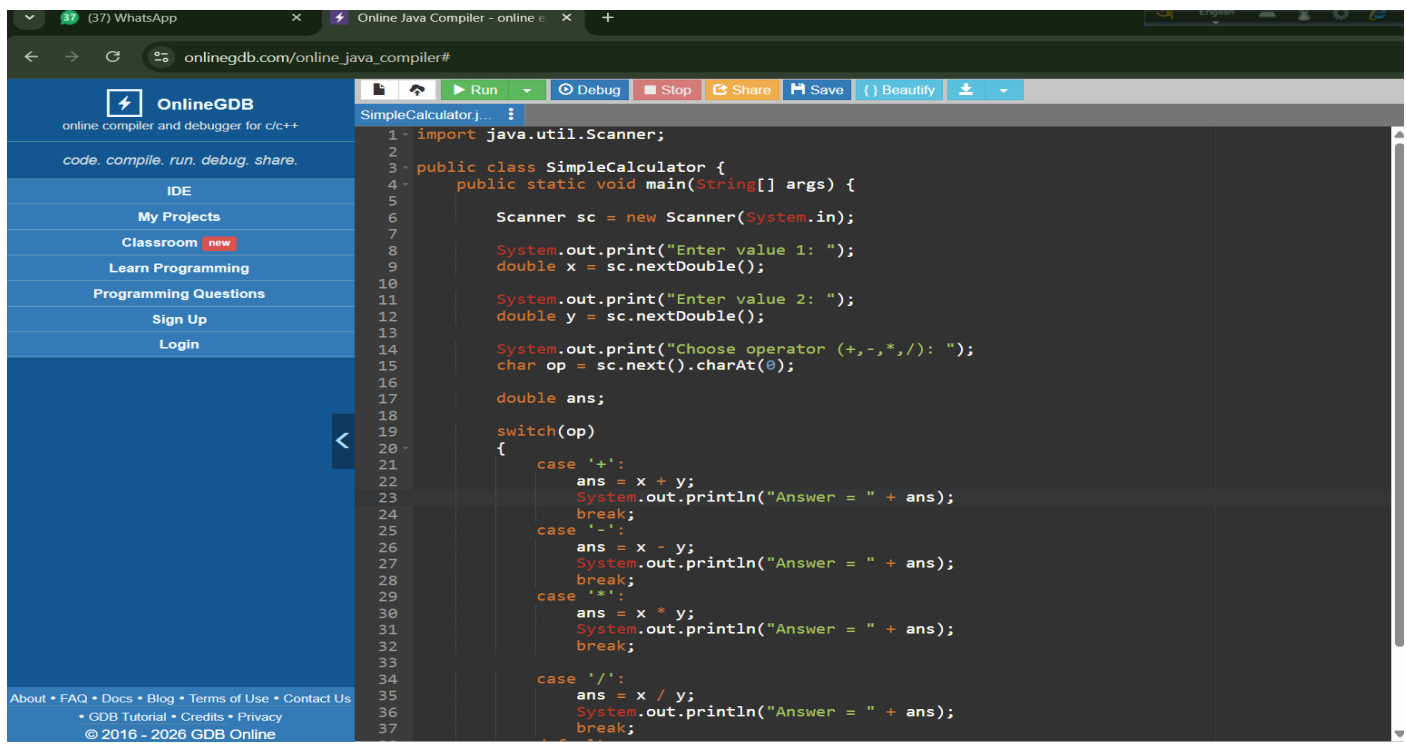
```
System.out.print("Choose operator (+,-,*,/): ");

char op = sc.next().charAt(0);


double ans;

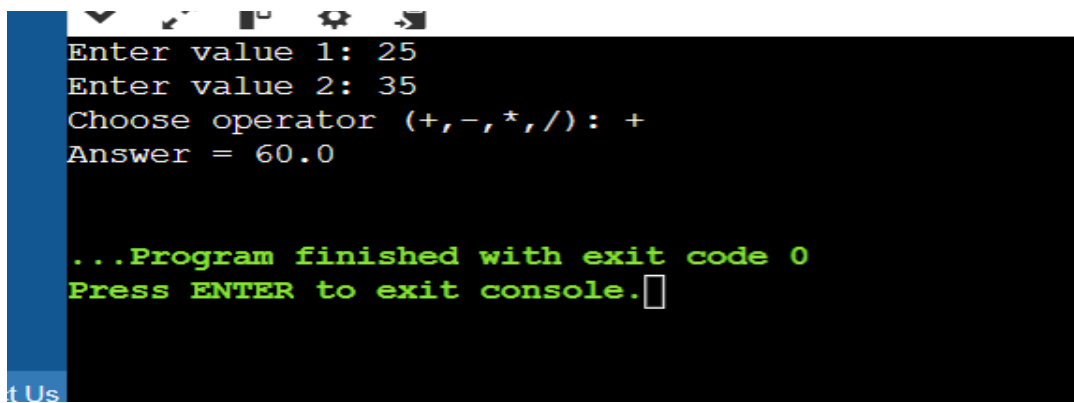

switch(op)
{
    case '+':
        ans = x + y;
        System.out.println("Answer = " + ans);
        break;
    case '-':
        ans = x - y;
        System.out.println("Answer = " + ans);
        break;
    case '*':
        ans = x * y;
        System.out.println("Answer = " + ans);
        break;
    case '/':
        ans = x / y;
        System.out.println("Answer = " + ans);
        break;
    default:
        System.out.println("Wrong operator");
}
}
```

Input:



```
1 import java.util.Scanner;
2
3 public class SimpleCalculator {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter value 1: ");
9         double x = sc.nextDouble();
10
11        System.out.print("Enter value 2: ");
12        double y = sc.nextDouble();
13
14        System.out.print("Choose operator (+,-,*,/): ");
15        char op = sc.next().charAt(0);
16
17        double ans;
18
19        switch(op)
20        {
21            case '+':
22                ans = x + y;
23                System.out.println("Answer = " + ans);
24                break;
25            case '-':
26                ans = x - y;
27                System.out.println("Answer = " + ans);
28                break;
29            case '*':
30                ans = x * y;
31                System.out.println("Answer = " + ans);
32                break;
33            case '/':
34                ans = x / y;
35                System.out.println("Answer = " + ans);
36                break;
37        }
```

Output:



```
Enter value 1: 25
Enter value 2: 35
Choose operator (+,-,*,/): +
Answer = 60.0

...Program finished with exit code 0
Press ENTER to exit console.
```

Discussion & Conclusion: In this experiment, a Java calculator program was developed using the switch-case statement. The program takes two numbers and an operator (such as +, -, *, /) as input from the user. Based on the selected operator, the corresponding case is executed to perform the calculation. The switch-case structure makes the program more organized and efficient compared to multiple if-else statements. This experiment helps to understand how decision-making and multiple conditions are handled in Java. The experiment successfully demonstrates the use of the switch-case statement in Java to perform basic arithmetic operations. It enhances understanding of control structures and user interaction in programming. This concept is useful for building more advanced applications like calculators and menu-driven programs.